CHAPTER 6

TESTING OF HYPOTHESES

CHAPTER ESSENCE AT A GLANCE:
Master hypothesis testing fundamentals: Null and Alternative hypotheses, Type I and Type II errors, level of significance (alpha), power of a test, p-value, and critical decision regions.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Null vs Alternative Hypothesis:** Null hypothesis H0 represents the hypothesis of no difference or status quo, set up for possible rejection; Alternative hypothesis H1 represents the claim being researched.

• **Type I and Type II Errors:** Type I Error is rejecting H0 when H0 is actually true (probability = alpha, level of significance); Type II Error is failing to reject H0 when H0 is false (probability = beta).

• **Power of a Test:** The probability of rejecting a false null hypothesis, defined as Power = 1 - beta; testing seeks to maximize power for a fixed significance level alpha.

• **Critical Region and p-Value:** The critical region (rejection region) comprises sample values leading to rejection of H0; p-value is the probability of observing results as extreme as sample data under H0 (reject H0 if p-value <= alpha).

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ True}), \quad \beta = P(\text{Accept } H_0 \mid H_0 \text{ False})$$

Bivariate Decision Risk Matrix in Hypothesis Testing

$$\text{Power of a Test} = 1 - \beta = P(\text{Reject } H_0 \mid H_0 \text{ is False})$$

Sensitivity Measure to Detect Genuine Parameter Shifts

$$\text{Decision Rule} \iff |Z_{\text{calc}}| > Z_\alpha \rightarrow \text{Reject } H_0 \text{ at Level } \alpha$$

Rejection Condition Relative to Critical Threshold

$$p\text{-Value Criterion} \iff p \leq \alpha \rightarrow \text{Reject } H_0 \quad [\text{Statistically Significant}]$$

Continuous Evidential Decision Metric

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Failing to reject H0 does NOT prove H0 is true; it merely indicates there is insufficient sample evidence to reject it (verdict is 'not guilty', not 'innocent').

• **Alert:** Level of significance alpha is the maximum probability of committing Type I error; commonly set at 5 percent (0.05) or 1 percent (0.01).

• **Alert:** A two-tailed test splits alpha into two equal tails (alpha divided by 2 each); a one-tailed test concentrates alpha entirely in one tail.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Null Hypothesis (H0) and Alternative Hypothesis (H1) with examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Null Hypothesis (H0):** A statement of no effect, no difference, or status quo, formulated specifically to test whether sample evidence can disprove it (e.g. H0: mu = 50). **[1 Mark]**

• **Alternative Hypothesis (H1):** The operational hypothesis that contradicts H0, accepted when sample evidence provides sufficient grounds to reject H0 (e.g. H1: mu != 50 or H1: mu > 50). **[1 Mark]**

2. Explain Type I Error and Type II Error with a decision table. Define Power of a Test. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Type I Error (alpha):** Rejecting the null hypothesis H0 when it is actually true. The probability of Type I error is termed level of significance alpha. **[1.5 Marks]**

• **Type II Error (beta):** Accepting (failing to reject) the null hypothesis H0 when it is actually false. The probability of Type II error is denoted by beta. **[1.5 Marks]**

• **Power of a Test:** Defined as 1 - beta = P(Reject H0 when H0 is false); it measures the test's ability to correctly detect a false null hypothesis. **[1 Mark]**

3. Distinguish between a Two-Tailed Test and a One-Tailed Test. Give one example of each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Two-Tailed Test:** The critical rejection region is divided equally between both tails of the sampling distribution (alpha divided by 2 in each tail). Used when H1 is non-directional (e.g. H1: mu != mu0; testing if a new machine changes average package weight). **[1.5 Marks]**

• **One-Tailed Test:** The entire critical rejection region of size alpha is placed entirely in one tail (left or right). Used when H1 specifies a directional change (e.g. H1: mu > mu0; testing if a training program increases sales). **[1.5 Marks]**

4. Outline the five basic steps involved in testing a statistical hypothesis. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Five Sequential Steps:**
1. Formulate Null Hypothesis H0 and Alternative Hypothesis H1.
2. Choose the level of significance α (e.g. 0.05 or 0.01). **[1 Mark]**

• 3. Select the appropriate test statistic (Z, t, F, Chi-Square) and identify its sampling distribution.
4. Determine the critical value and define the critical rejection region. **[1 Mark]**

• 5. Compute the calculated test statistic from sample data and make the statistical decision (reject H0 if calculated value falls in critical region). **[1 Mark]**

5. What is meant by the p-value of a statistical test? How is it used to make a decision? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** The p-value is the lowest significance level at which the observed sample result would lead to rejection of the null hypothesis H0. **[1 Mark]**

• **Decision Rule:** If p-value <= alpha, reject H0 (statistically significant); if p-value > alpha, fail to reject H0. **[1 Mark]**

6. State the critical values of Z for a two-tailed test at: (a) 5 percent level of significance, (b) 1 percent level of significance. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) At 5 percent level (alpha = 0.05):** Critical Z = +- 1.96. **[1 Mark]**

• **(b) At 1 percent level (alpha = 0.01):** Critical Z = +- 2.58. **[1 Mark]**